

Vedant Subhash Phad

B.Tech - Chemical Science and Technology
Minor in Biotechnology
Indian Institute Of Technology, Guwahati

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EDUCATION

Degree/Certificate	Institute/Board	GPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology, Guwahati	8.45/10	2021-2025
B.Tech. Minor	Indian Institute of Technology, Guwahati	9.2/10	2022-2025
Senior Secondary	HSC Board	96.33%	2021
Secondary	ICSE Board	96.2%	2019

PUBLICATIONS

- Das, J., **Vedant Phad**, Dhabal, D.* and Manna, U.* **Facilitating Electrochemical Overall Water Splitting Through Tailoring Halloysite Nanoclay-Based Nanoarchitectonics of Ionic Environment and Gas-Bubble Wettability.** *Advanced Materials.* (in press)
- Giri, P., Borbora, A., Sarkar, D., Dutta, S., Weible, A. H., Sarma, H. **Vedant Phad**, Mohanta, A., Wang, X.* and Manna, U.* **Coupling programmable shape morphing and solvent-fueled propulsion in a soft bicontinuous composite.** *Nature Communications.* (under review)
- **Vedant Phad**, Farhat, Z., Prasad, A. and Kumar, S.* (2024). **Newer approaches for viral vaccine development.** In *Bioreactor Design Concepts for Viral Vaccine Production* (pp. 389-399). Academic Press. (**ISBN: 9780443153785**)
- **Vedant Phad** and Dhabal, D.*, **Reparametrized OPLS Force Field of Diiodomethane for Surface Free Energy Calculations** (under preparation).
- Debnath, I., **Vedant Phad**, Dhabal, D.* and Mahata, K.* **Exploring Supramolecular Interactions and Aggregate Formation of peri-Naphthoisatogen Using Molecular Dynamic Simulations** (under preparation).
- **Vedant Phad** and Dhabal, D.*, **On the Accuracy of ML-BOP Model in Reproducing the Properties of High-Pressure Ices** (under preparation).
- Daimari, D. D., Das, J., **Vedant Phad**, Manna, U.* and Dhabal, D.* **Investigating the mechanism of anti-icing and de-icing by ionic liquids** (under preparation).

RESEARCH EXPERIENCE

- **Bio-Inspired Polymeric Materials Lab** May 2025 - present
Supervisor: Prof. Uttam Manna Department of Chemistry, IIT Guwahati
 - Conducted experimental electrochemistry for water splitting, focusing on the hydrogen and oxygen evolution reactions (HER/OER) on polymer-functionalized electrodes.
 - Designed and investigated hydrogels infused with hydrophobic organic molecules, imparting shape-morphing and shape-memory characteristics in addition to mechanical stability.
- **Computational Material Science and Biomaterials Lab** Feb. 2024 - present
Supervisor: Dr. Debdas Dhabal Department of Chemistry, IIT Guwahati
 - Calculated contact angle of water droplet and Diiodomethane droplet on chemically modified surfaces using MD simulations with the objective to enhance hydrogen evolution reaction (HER). (*Co-PI - Dr. Uttam Manna*)
 - Performed MD simulations on peri-Naphthoisatogen (active component in the application of fluorescence thermometers) substituted with long alkyl chains in THF and water solvent system using **Gromacs**. The simulation results explained supramolecular interactions and aggregate formation. (*Co-PI - Dr. Kingsuk Mahata*)
 - Validated the performance of ML-BOP coarse-grained model to simulate high-pressure ice phases, analyzing its accuracy in reproducing thermodynamic and structural properties compared to experimental data.
- **Immunology and Microbiology Lab - Virtual Labs** Feb. 2023 - Jan. 2024
Prof. Sachin Kumar/Virtual Labs/Ministry of Education <https://im-iitg.vlabs.ac.in/>
 - Performed experiments and analyzed results: Flow cytometry of immune cells, MIC determination of antibiotics, cloning and expression of GFP sequence in expression vector.
 - Documented experimental procedures and uploaded data on the portal.

CONFERENCES ATTENDED

- **SoPhyc (Society of Physical Chemistry)** Oct. 2025
Department of Chemistry, IIT Patna
 - Presented a poster on my work on "Facilitating electrochemical overall water splitting through tailoring ionic environment and gas-bubble wettability."

- I highlighted the impact of synergistic association of bubble repellence and selective ionic environments on water splitting performance.

- **FICS (Frontiers in Chemical Sciences)**

Dec. 2024

Department of Chemistry, IIT Guwahati

- Presented a poster on my work on "On the Accuracy of ML-BOP Model in Reproducing the Properties of High-Pressure Ices".
- I addressed ML-BOP's computational efficiency and its ability to reproduce key properties of high-pressure ices while identifying areas for further improvement, such as limitations in lattice energy predictions.

TECHNICAL SKILLS

- **Molecular Docking softwares:** Pymol, AutoDock Vina
- **MD simulation softwares:** Gromacs, LAMMPS*, VMD
- **DFT softwares:** GaussView
- **Processes studied:** Gel Electrophoresis, Western Blotting, PCR, MIC Assay
- **Design:** Blender*, BioRender, Canva, Microsoft Powerpoint
- **Operating Systems:** Linux, Windows
- **Documentation Softwares:** Microsoft Word, EndNote
- **Miscellaneous:** Fortran, MATLAB*, Latex*

* Elementary proficiency

KEY COURSES TAKEN

- **Chemistry:** Introduction to Quantum Chemistry, Group Theory, Application of Nanomaterials, Computational Chemistry, Polymer Chemistry, Medicinal Chemistry, Spectroscopic techniques, Drug discovery and development
- **Biotechnology:** Biochemistry, Molecular Biotechnology, Cellular Biotechnology, Bioanalytical techniques and Bioinformatics, Fluorescence techniques in biotechnology
- **Chemical Engineering:** Chemical Process Calculations, Fluid Mechanics
- **Computer Science:** Introduction to C, Computer aided optimization
- **Mathematics:** Basic Calculus, Introductory Mathematical Economics
- **Online Certified Courses:** MATLAB Onramp, Optimization Onramp (MathWorks)

LEADERSHIP ROLES

- **General Secretary, CheSTA**, student body of Chemistry department Aug. 2024 - May 2025
 - Led departmental student body initiatives to strengthen alumni relations, improve placement outreach, and enhance visibility of CST students through branding, brochures, and website revamp.
 - Built a structured student team and launched academic, mentorship, and well-being programs, including alumni talks, and student-faculty communication platforms.

ACHIEVEMENTS

- **GATE Biotechnology and Life Science**, Qualified with AIR - 804 and 271 respectively. 2024
- **Joint Entrance Examination Main**, Achieved 97.42th Percentile among 1.04 million candidates across India. 2021
- **Joint Entrance Examination Advanced**, Secured position in top 2.07% among 0.14 million candidates appearing for the examination. 2021

EXTRACURRICULARS

- **Volunteer at National Service Scheme**, helping in social service activities Jan. 2023 - Present
- **Organizer**, Techniche -annual techno-management festival of IIT Guwahati, Corporate Module Sept. 2022
- **Performed in a Play (theatre)**, organised by Xpression, the Dramatics club of IIT Guwahati 22nd Feb. 2023
- **Performed in Street play**, conveyed impactful social messages through the performance 15th Aug. 2022
- **Music**, Love Playing Piano, Melodica, Harmonium and Tabla

REFERENCES

Dr. Uttam Manna

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Dr. Debdas Dhabal

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Dr. Sachin Kumar

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